AEROSENSE TEACHER TOOLKIT

Made by AeroSense · NTHU iGEM 2026

Teacher Toolkit

How a nose smells, and how an electronic nose smells:

a 40-minute class, with a game and an online platform.

1. About this kit

This kit lets a teacher run the class we taught at an elementary school science camp, without us in the room. We ran it with more than thirty pupils in grades 5 and 6. It has three parts: the lesson slides, a game, and an online learning platform.

Toolkit folder: open the shared folder

- **Lesson slides:** the Science Mini-Class, 26 slides. Open the slides
- **Game:** Odor Pixel Suite, played as a whole-class challenge. Open the game
- **Online platform:** the AeroSense Learning Lab, for follow-up learning. Open the Learning Lab
- **Worksheet:** for the drawing activity.

2. Before you start

- A projector or screen, and a computer to open the slides and the game.
- Paper and coloured pencils or crayons, and one worksheet per pupil.
- A few perfumes or scents for the smelling activity, sprayed onto paper strips. Ask about allergies and asthma first.
- Divide the class into teams before the game, so that each team can answer together.
- Open the game link once beforehand to check that it loads on your network.

3. The 40-minute lesson

Time	Activity	What happens
10 min	**Science Mini-Class**	Three questions in a row: How does a nose smell? How does an electronic nose smell? What does our lab do?
10 min	**Odor Pixel Suite challenge**	Questions are projected on a screen; each team answers at the same time by holding up 1, 2 or 3 fingers.
15 min	**Drawing and worksheet**	Pupils draw a cartoon nose or a cartoon fruit fly, then complete the worksheet.
5 min	**Wrap-up**	Short review of the three questions; collect drawings and worksheets.
40 min	**Total**	One class period, or half of a school morning if you add breaks.

4. Step by step

Step 1 · Science Mini-Class (10 min)

The lesson slides, “Teaching a machine to smell,” follow the three opening questions. The slides build up one idea at a time, so click through slowly and let the children guess before each answer appears. The timings below are a suggestion.

Question 1 · How does a nose smell? (about 4 min). Slides on “What is a smell?”: a smell is tiny flying molecules; the nose has many little locks; the right key fits the right lock and the brain knows “this is a banana.”

Interaction: smelling. Right after the “little locks” slide, let the children smell a few perfumes or scents and describe them. We suggest spraying onto paper strips rather than into the air, passing the strips around, asking about allergies and asthma first, and keeping the room ventilated.

Question 2 · How does an electronic nose smell? (about 3 min). Slides on the electronic nose: a row of small artificial noses, each reacting differently, like a jury voting together; then what it can do, such as guarding safety, helping diagnosis, airport screening and checking food.

Interaction: ask first. Before each use-case slide, ask “Where would a machine nose help?” and take two or three answers.

Question 3 · What does our lab do? (about 3 min). Slides “Nature is our teacher” (the dog’s nose and the fly’s brain) and “The machine’s nose gets tired too.” Use the grandmother’s-house example: you smell the food at first, then you stop noticing, so your nose has “got used to it.” An electronic nose also wears out with use, and our job is to teach it not to be fooled.

Step 2 · Odor Pixel Suite challenge (10 min)

Open the game on the teacher’s computer and project it. Instead of one child at one screen, show each question on the screen and let every team answer at the same time by holding up 1, 2 or 3 fingers. Count the fingers, then reveal the answer and say why. Game link

Odor Pixel Suite has three short games. You can run the whole-class challenge as above, or let pupils play on their own devices.

- **Pattern Recognition (10–15 min):** identify an odor from its pattern. A smell is a pattern across many receptors, not one reading.
- **Odor Maze (8–20 min):** find the way out using odor clues.
- **Odor Fingerprint Mixer (15–25 min):** mix two odors and work out the mixture.

Step 3 · Drawing and worksheet (15 min)

Ask the children to draw a cartoon nose or a cartoon fruit fly, “the electronic nose and the fruit fly as you imagine them,” and then complete the worksheet. Drawings show what a child took away in their own terms. Collect them at the end.

Step 4 · Wrap-up (5 min)

Go back over the three opening questions and let a few children answer them in their own words.

5. Using the online platform

The Learning Lab has five modules of about 8 minutes each, about 40 minutes in total: Discover, Question, Build, Decode and Decide. Modules can be taken in any order, but they are designed to follow the logic of the project. Learning Lab link

- **In class:** project one module and go through it together, then discuss. Printable discussion prompts for educators are on the platform.
- **As homework:** assign one or two modules. Older students can go through all five.
- **Check questions:** each module ends with three check questions, so learners can test themselves.

6. After the class

- Keep the drawings and worksheets. They show what the children understood and where they got stuck.
- Before sharing any child's drawing or photo outside the school, get consent from the school and from guardians.
- Send us your drawings or feedback if you wish, so they can join our Art Gallery.

7. Contact

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教師教案包

鼻子怎麼聞，電子鼻怎麼聞：一堂 40 分鐘的課，搭配遊戲與線上學習平台。

1. 這份教案包是什麼

這份教案包讓老師不需要我們到場，也能帶出我們在國小科普營隊上的那堂課。我們當時的對象是五、六年級三十多位學生。內容分三部分：課程簡報、遊戲，以及線上學習平台。

教案包資料夾：開啟共用資料夾

- **課程簡報：**科普小學堂，共 26 頁。開啟簡報
- **遊戲：**Odor Pixel Suite，以全班闖關方式進行。開啟遊戲
- **線上平台：**AeroSense Learning Lab，供課後延伸學習。開啟 Learning Lab
- **學習單：**畫畫活動使用。

2. 開始前準備

- 投影機或布幕，以及一台開啟簡報與遊戲的電腦。
- 紙、色鉛筆或蠟筆，每位學生一份學習單。
- 幾種香水或香味，供聞香活動使用，噴在紙條上。先詢問有無過敏或氣喘。
- 遊戲前先把全班分組，讓各隊可以一起作答。
- 課前先開啟一次遊戲連結，確認在學校網路上可以使用。

3. 一堂 40 分鐘的課

時間	活動	做法
10 min	**科普小學堂**	鼻子怎麼聞 → 電子鼻怎麼聞 → 實驗室在做什麼
10 min	**Odor Pixel Suite 闖關**	題目投影，各隊同時舉手指比 1、2、3 作答
15 min	**畫畫＋學習單**	畫卡通鼻子或卡通果蠅，再完成學習單
5 min	**收尾**	回顧三個問題，收回圖畫與學習單
40 min	**合計**	一堂課的長度；加上休息可排成半天

4. 帶課步驟

步驟一 · 科普小學堂（10 分鐘）

課程簡報「教機器學會聞味道！」依三個開場問題進行。簡報一次出現一個概念，請慢慢點，並在每個答案出現前先讓孩子猜。下面的時間分配僅供參考。

問題一 · 鼻子怎麼聞？（約 4 分鐘）。簡報「味道是什麼？」：味道是會飛的小分子；鼻子裡有很多小鎖；鑰匙插進對的鎖，大腦就知道「這是香蕉」。

互動：聞香味。在「小鎖」那一頁之後，讓孩子聞幾種香水或香味，並說說聞起來像什麼。建議噴在紙條上傳著聞，不要直接噴在空氣中；先詢問有無過敏或氣喘，並保持教室通風。

問題二 · 電子鼻怎麼聞？（約 3 分鐘）。簡報「電子鼻」：一排小小的人工鼻子，每個反應都不同，像評審團一起投票；接著介紹它能做的事，例如守護安全、幫忙看病、機場把關、檢查食物。

互動：先問再答。每個應用頁出現前，先問「機器鼻子可以在哪裡幫上忙？」，聽兩三個答案。

問題三 · 我們實驗室在做什麼？（約 3 分鐘）。簡報「我們的老師是大自然」（狗鼻子與蒼蠅的腦）與「機器的鼻子也會累」。用「走進阿嬤家」的例子：一開始聞到飯菜香，過一陣子就聞不到了，是你的鼻子「習慣」了。電子鼻用久了也會累，我們的任務是教它不要被騙。

步驟二 · Odor Pixel Suite 闖關（10 分鐘）

在老師的電腦開啟遊戲並投影。不是一個孩子對一台螢幕，而是把題目放在螢幕上，讓每一隊同時舉手指比 1、2 或 3 作答。數完手指後再公布答案並說明原因。遊戲連結

Odor Pixel Suite 共有三款短遊戲。你可以用上面的全班闖關方式，也可以讓學生用自己的裝置玩。

- **圖樣辨識（10–15 分鐘）：**從圖樣辨認氣味。氣味是多個受體的圖樣，不是單一數值。
- **氣味迷宮（8–20 分鐘）：**用氣味線索找到出口。
- **氣味指紋混合器（15–25 分鐘）：**混合兩種氣味並推斷成分。

步驟三 · 畫畫與學習單（15 分鐘）

請孩子畫一個卡通鼻子或卡通果蠅，也就是「你心目中的電子鼻和果蠅」，再完成學習單。圖畫能看出孩子用自己的方式帶走了什麼。結束時收回。

步驟四 · 收尾（5 分鐘）

回頭看開場的三個問題，請幾位孩子用自己的話回答。

5. 線上學習平台怎麼用

Learning Lab 有五個模組，每個約 8 分鐘，合計約 40 分鐘：Discover、Question、Build、Decode、Decide。模組可以任意順序進行，但設計上依專案的邏輯排列。Learning Lab 連結

- **課堂上：**投影一個模組，全班一起進行並討論。平台上有可列印的教師討論題。
- **作為回家作業：**指定一到兩個模組。年紀較大的學生可以完整做完五個。
- **檢核題：**每個模組最後有三題，讓學習者自我檢測。

6. 課後

- 保留圖畫與學習單，可以看出孩子理解了什麼、卡在哪裡。
- 要在學校以外公開任何孩子的圖畫或照片前，請先取得學校與家長同意。
- 歡迎寄給我們圖畫或回饋，讓它們加入我們的 Art Gallery。

7. 聯絡方式

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